

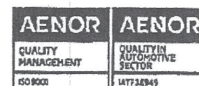
Sidenor

#508271

MILL TEST CERTIFICATE

Azkoitia Plant

ISO 9001; IATF 16949; ISO 14001; ISO 45001; ISO 50001; ISO 27001 Y TISAX
 SIDENOR ACEROS ESPECIALES S.L.U.
 Cr. Zumarraga 8/n. 20720 Azkoitia (Guipuzcoa)
 TLF. 943025222 FAX 943025252
 www.sidenor.com



R817388

Product Made in Spain

CUSTOMER:MAGELLAN CORPORATION	WORKS REFERENCE:2319371
REFERENCE:216438	SALES ORDER:363343-1
PRODUCT NR:	HEAT NUMBER:645158
MASTER REFERENCE:287805	ROLLED:20.05.2024

REQUIRED PRODUCT

NITRIDING 135 MOD ROUND BARS TURNED Q+T+STRESS RELIEVED 1.1250"

ASTM A-108 .(") alloy 0,28<C<=0,55 (U,R,N) 12'/12'2" NORMAL

EXPEDITION

DELIVERY:0080843565 WEIGHT (KG):8.192

BUNDLES:4

UNITS:449

MADE ACCORDING TO

ASTM A108 18 01.11.2018; ASTM A255 -20A 01.11.2020; ASTM A29-A29M -23 01.07.2023; ASTM A370 -22 11.2022
 ASTM A751 21 01.11.2021; ASTM E112 -13(R2021) 01.09.2021; ASTM E381 -22 06.2022
 CASTLE METALS N13500-01 17 01.06.2020; EN 10204 - OCT. 2004 3.1; MAGELLAN MIC2016 1 30.01.2012
 MIL 86709A-NOT2 - 12.12.1994; SAE-AMS 2301 L 05.2020; SAE-AMS 6470 P 08.2018; SAE-AMS 6472 G 08.2018
 SAE-AMS 66709 A SEPT. 2002

CHEMICAL ANALYSIS OF HEAT

U: % HEAT NUMBER:645158

	C	Mn	Si	P	S	Cr	Ni	Mo	Cu	Al
Min.	0,380	0,500	0,200			1,400		0,300		0,950
Max.	0,430	0,700	0,400	0,025	0,025	1,800	0,250	0,400	0,350	1,300
Cer.	0,395	0,669	0,216	0,008	0,001	1,578	0,178	0,355	0,211	1,018
D.I.ASTM A255 (")	8,407									

JOMINY HARDENABILITY

HARDNESS U.:HRC NORM.TEMP.:1.800-10/10°F

8/16" 12/16"

Min. 50,0 45,0

Max.

Cer. 56,0 55,5

JOMINY Standard (ASTM A255-20A01.11.2020)

Temperature of (Austenitizing 1.700-10/10°F):Austenitizing 1.700-10/10°F

INCLUSIONS (MICROINCLUSIONS)

Microinclusions std. (ASTM E4518A(R2023)01.11.2023); Type/method(A); A(t):0,9; A(h):0,5; B(t):2,5; B(h):2
C(t):0; C(h):0; D(t):1,5; D(h):1

INCLUSIONS (MACROINCLUSIONS)

Standard (SAE-AMS 2301 L 05.2020); Severity(1):Mean value 0,17; Severity(2):Individual value 0,26
Frequency(1):Mean value 0,21; Frequency(2):Individual value 0,31

MECHANICAL PROPERTIES AS SUPPLIED (CONDITIONS)

Temperature of:(1):Quenching 1.706°F; Time:(1):1,2hours; Cooling:(1):ACETONE

Temperature media at the start:(1):122°F; Temperature media at the finish:(1):129°F

Temperature of:(2):Tempering 1.274°F; Time:(2):1,2hours; Cooling:(2):AIRE

Temperature of:(3):Stress Relieved 1.130°F; Time:(3):6hours; Cooling:(3):AIRE

Furnaces survey and calibration:OK

MECHANICAL PROPERTIES AS SUPPLIED (TEST)

Specimen Test location:Nucleus

Tensile std. (ASTM A370-2211.2022); Tensile dir.:Longitudinal

Ts(>=112KSI):130KSI; Ys(1) (Rp(0,2%) >=90KSI):Rp(0,2%) 106KSI; Ys(2) (0,02% KSI):0,02% 100KSI; Ys/Ts:0,81%

El.(4d >=16%):4d 22,7%; Z(>=50%):64,3%

Hardness Standard (ASTM E10 -18 01.07.2018); Hardness at 1/2 radius (241/285HB):275HB; Core hardness (HB):272HB

TECHNOLOGY & QUALITY CERTIFIES THAT THE PRODUCT FULL FILLS THE ORDER'S SHEET

APPROVED BY:Manuel Dávila Piris

SIGN:

DATE:12.09.2024

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REF.:2001755350000

Analyst of Quality certificates

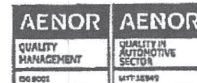
Customer:BLUE CHIP MACHINE CO | Ord #:T534848 | Part #:508271 | Cust PO #:27005

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PO 001854
PL 309-150-06-MC001K-R0

MILL TEST CERTIFICATE



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ADDITIONAL TESTS (SUPPLY CONDITIONS)

Decarburization:0; Grain size std.(ASTM E112-13(R2021)01.09.2021); Grain size:7
Macroetching std.(ASTM E381-2206.2022); Macroetching(1):S 1; Macroetching(2):R 1; Macroetching(3):C 1
Radiactivity:FREE <= 0,2µSV/H

NON DESTRUCTIVE TESTS

Surface defects std.(ASTM E709-2101.06.2021)
U.T. standard(ASTM A388-A388M -19 01.09.2019); ULTRASONIC INSPECTION 100% : O.K.; CRACKS CONTROL 100%:OK

ADDITIONAL INFORMATION

Reduction ratio:304,58; Initial format, reduct. ratio, dimen:Ingot; Magnetism:<= 10Gauss
Add. info(Material 100% melted and poured in Spain); MIL-S-6709 Condition "F4; Free of Magnetism
UNS-K24065; Acc to SAE-AMS 6472, except pirometry Acc to SAE-AMS 2750

Material melted and manufactured 100% in Spain through the Electric Arc Furnace (EAF) and Vacuum Degassing route (VOD).
Steel not exposed to Mercury, or to any other metal alloy that is liquid, at ambient temperatures during processing or while in Sidenor's possession.
Material dimensionally controlled, visually inspected, and free from radioactivity (<0,1Bq/g concerning a Co-60).
Steel products were not repaired by welding.
100% anti mix test performed.

TECHNOLOGY & QUALITY CERTIFIES THAT THE PRODUCT FULL FILLS THE ORDER'S SPECIFICATIONS

APPROVED BY:Manuel Dávila Piris

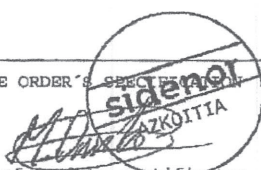
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Analyst of Quality certificates



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MAGELLAN
CORPORATION

a SeaTrade company

1650 Lake Cook Road Suite 300A Deerfield, IL 60015 USA

Telephone: (847) 205-1155

MOR: 708419

SR: 20290317

Packing List / Certificate of Conformance

Sold To: EMJ-Schaumburg

Producing Mill: Sidenor Aceros Especiales S.L.U.

Customer PO / Item	Bundle #	Description	Heat # / Lot #	Pieces	Weight Pounds
P329409-423 / 508271	991879	NITR-CFOTS-8RRD-1.1250	645158 / 2319371	124	4.978

Total
124 - 4.978